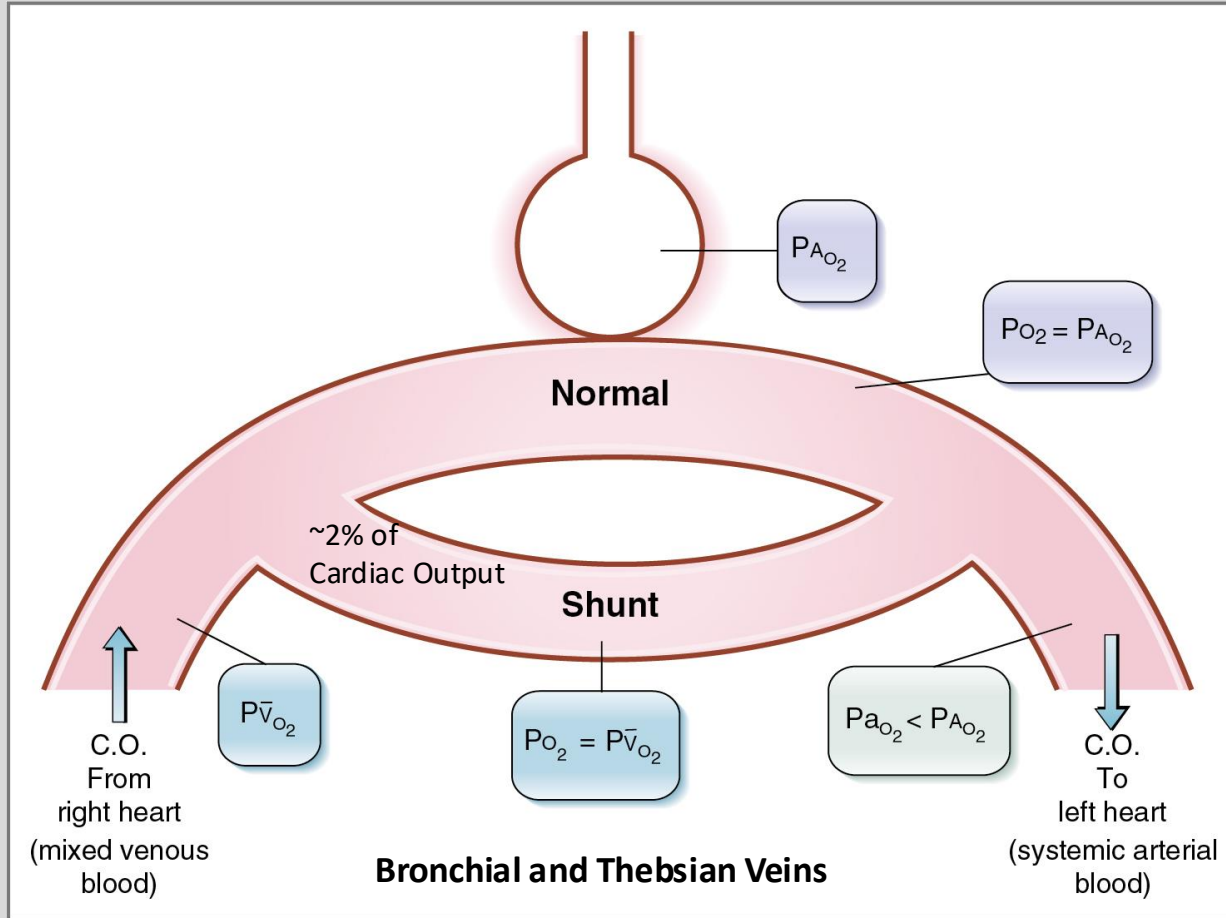


Objectives:

- Compare & contrast physiologic and non-physiologic shunts based on their causes and consequences.
- Explain how the various kinds of shunts respond to supplemental oxygen.
- Compare & contrast focal vs. global hypoxia.
- Compare & contrast the effect of lung volume on alveolar and extra-alveolar vessels.
- Review & recall select pulmonary vasoconstrictors/dilators & airway constrictors.
- Describe the gravitational effects of pulmonary blood flow on the $\dot{V}/\dot{Q}$ at the apex, center and base of the upright lungs and the physiologic challenges that ensue.
- Describe how those differences across the lungs challenge gas exchange.
- Describe how compensatory bronchoconstriction & compensatory hypoxic vasoconstriction can remedy $\dot{V}/\dot{Q}$ mismatches.
- Compare and contrast the major causes of $\dot{V}/\dot{Q}$ defects and the consequences on gas exchange (hypoxemia, hypercarbia).

Physiologic Right – to – Left Shunts



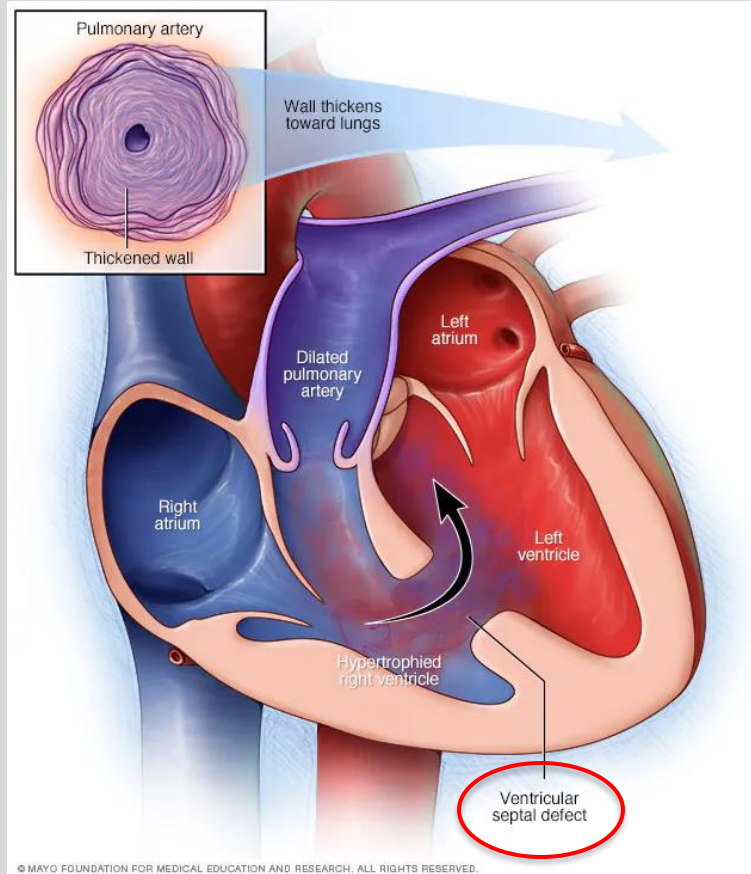
Are there consequences
for Pa_{O_2} ?

How about Pa_{CO_2} ?

Hint: Consider high
sensitivity of CO_2 receptors

Example of a Non-Physiologic Right-to-Left Shunt

Hypoxemia Always Occurs – Cannot be Corrected with Breathing O₂



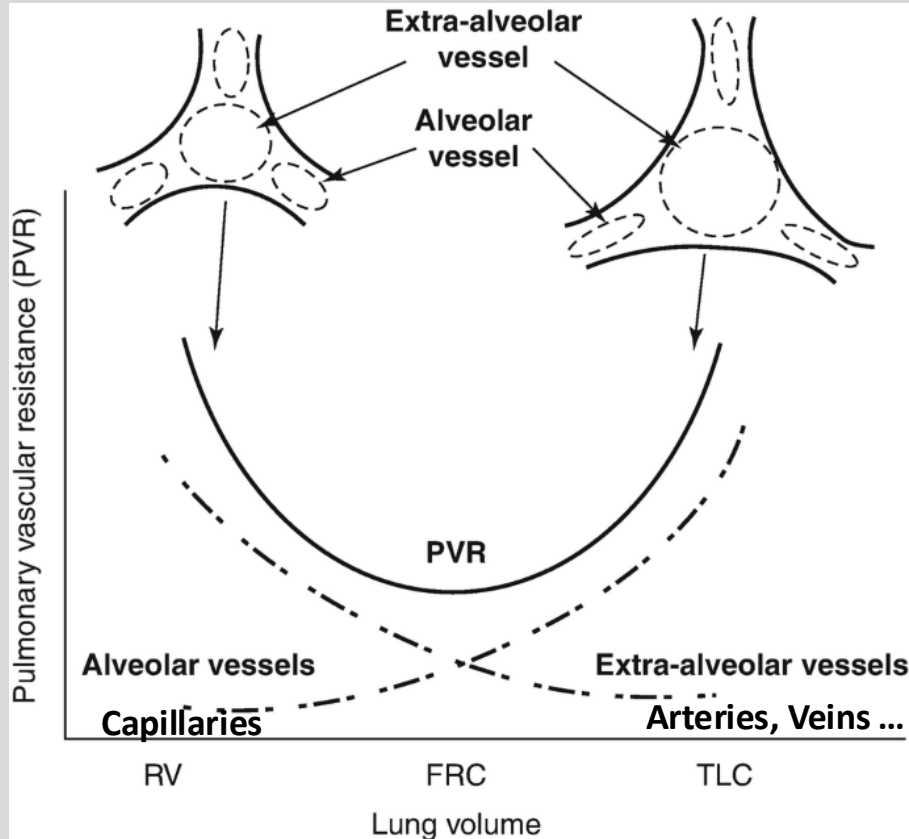
Eisenmenger Syndrome and others

Shunt fraction equation:

$$\frac{Q_s}{Q_T}$$

Q_S = Shunted Blood; Q_T = Cardiac Output

Lung Volume & Pulmonary Vascular Resistance - Reminder



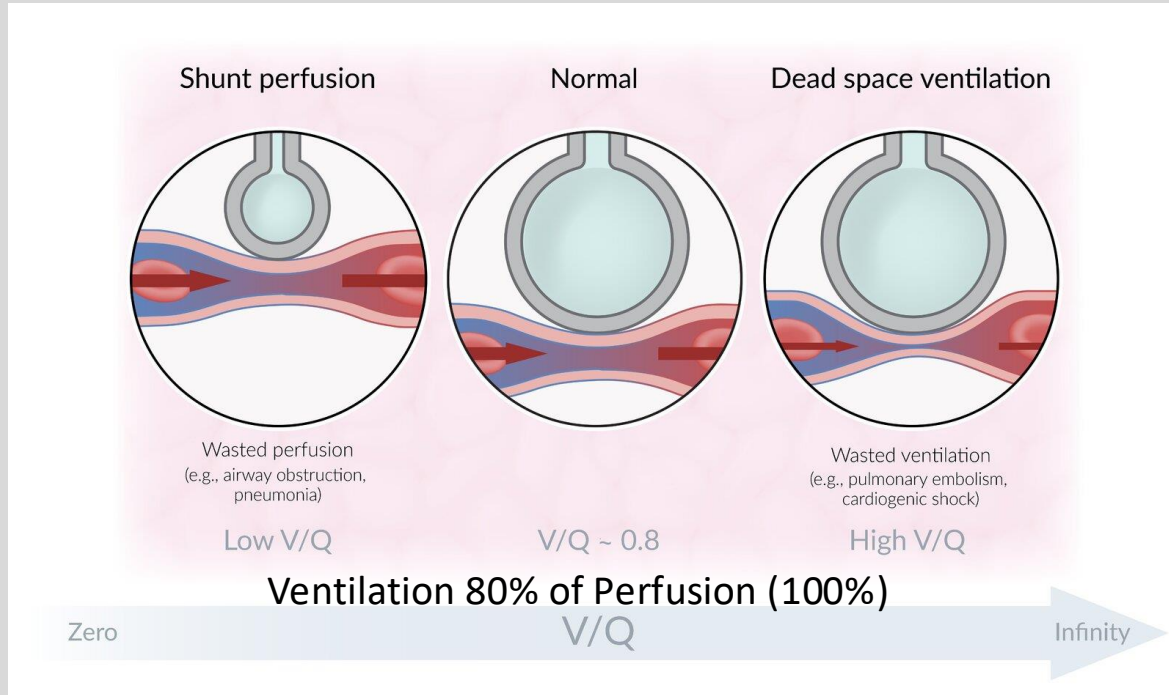
Lung volume effects **extra-alveolar** and **alveolar** blood vessels in **opposite ways**

Ventilation/Perfusion Ratios

$\dot{V}/\dot{Q}$ Matching

Two Mechanisms:

Compensatory **Bronchoconstriction** & Compensatory **Hypoxic Vasoconstriction**



A 32-year-old female is admitted to the intensive care unit with acute respiratory distress syndrome (ARDS) secondary to severe atypical pneumonia. Her chest X-ray shows diffuse, bilateral alveolar infiltrates and extensive fluid-filled, collapsed alveoli (atelectasis), creating a massive physiological right-to-left shunt. The resident on duty places the patient on 100% supplemental oxygen via a non-rebreather mask, but her arterial oxygen saturation (S_aO_2) remains critically low at 82%. Which of the following best explains why her hypoxemia is highly resistant to supplemental oxygen therapy?

- A. The high concentration of oxygen causes widespread pulmonary vasodilation, which increases the shunt fraction.
- B. Supplemental oxygen decreases the affinity of hemoglobin for any remaining oxygen in the systemic circulation.
- C. The inspired oxygen cannot reach the blood flowing through the collapsed, unventilated alveoli.
- D. Oxygen gas is highly insoluble in blood plasma, preventing any increase in dissolved oxygen content.
- E. The elevated oxygen levels inhibit the enzyme carbonic anhydrase, blocking normal gas transit.